

Sound Becomes a Signal

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Pictures Become Numbers

Without opening the old lesson, explain one key idea from Pictures Become Numbers. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Sensors Are Messengers

Without opening the old lesson, explain one key idea from Sensors Are Messengers. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Inspect what was discarded for Sound as sampled information. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Sound as sampled information. Proposed explanation: “Five samples make five gaps”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Construct a new hidden change

A record keeps values 0 at times 0,3,6 milliseconds. Construct two seven-sample sequences for every whole-millisecond time from 0 through 6 that agree with those retained values but differ at omitted times. Explain why the retained record cannot prove that nothing changed between measurements.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A whisper recorded close to a microphone has larger sample amplitudes than a shout recorded far away. Does the recording prove the whisper was louder at its source? Propose one controlled follow-up.

6. Solve the given case

A microphone records quiet and loud claps at the same distance and settings.

What waveform difference is expected? Does it reveal a spoken word?

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Two recordings contain the same word: one quiet and one loud. Which device receives the sound? Does a larger wave automatically mean a different word? Explain.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 6 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

If the sorter uses sound, specify the exact sound pattern needed and test whether classroom noise makes that input unreliable.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A whisper recorded close to a microphone has larger sample amplitudes than a shout recorded far away. Does the recording prove the whisper was louder at its source? Propose one controlled follow-up.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Read samples and gaps

Five samples have values 0,3,0,1,0 at times 0,2,4,6,8 milliseconds. State the sample count, interval and elapsed time from first to last. Explain why there are four gaps.

My reasoning and evidence

13. W2 Inspect what was discarded

Retain only times 0,4,8 from W1. Write the retained values. Give one claim about the original signal that these retained values do not justify.

My reasoning and evidence

14. W3 Compare equal summaries

Sequences $A=0,2,0,2,0$ and $B=0,0,2,0,0$ use the same time interval. State their maximum values and one difference the maximum discards. Name information that would preserve that difference.

My reasoning and evidence

15. W4 Construct a new hidden change

A record keeps values 0 at times 0,3,6 milliseconds. Construct two seven-sample sequences for every whole-millisecond time from 0 through 6 that agree with those retained values but differ at omitted times. Explain why the retained record cannot prove that nothing changed between measurements.

My reasoning and evidence

16. W5 Check a speech pipeline

A recording says Set the timer for fourteen minutes. A transcript says forty minutes and the timer is set to 40. Name the first mismatch, propose a confirmation step and explain why the transcript cannot be treated as a complete copy of the recording.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Sound Becomes a Signal

A fictional microphone display records five samples during a short classroom tap.

Times are 0,1,2,3,4 ms in order.

Signal A: 0,3,0,1,0. Signal B: 0,1,0,3,0. Toy values are signed measurement units; these happen to be nonnegative.

AI ACADEMY / AI-1/6

Same peak, different sample timing

	0 ms	1 ms	2 ms	3 ms	4 ms
Signal A	0	3	0	1	0
Signal B	0	1	0	3	0

Read a cell using both its row and column labels.

Sound Becomes a Signal

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Timed sampling

Associate every measurement with its sampling time.



2. Signal A

Its largest sample 3 occurs at 1 ms.



3. Signal B

Its largest sample 3 occurs at 3 ms.



4. Maximum summary

Both maxima are 3, but the strongest recorded tap arrives at different times.

CHANGE ONE THING

Retain only samples at 0,2,4 ms.

Explain the changed result using the matching lesson.

Does the all-zero retained record prove the full signal was silent? Use the picture.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Record the sorter's sampling interval and retain relevant sequences when a decision depends on changes over time.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

Record the sorter's sampling interval and retain relevant sequences when a decision depends on changes over time.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional microphone display records five samples during a short classroom tap.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Both maxima are 3, but the strongest recorded tap arrives at different times.

1. Signal A
2. Signal B
3. Maximum summary
4. Timed sampling

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Maximum summary Both maxima are 3, but the strongest recorded tap arrives at different times. The sequences contain the same values in different orders. Keeping the maximum preserves peak recorded value while losing its timing. A graph should keep time on the horizontal axis; shuffling samples changes the recorded signal even if its maximum stays fixed.

Practice 2: explain the deciding evidence

Changed-case practice: Retain only samples at 0,2,4 ms. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. Both retained sequences are 0,0,0.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Both retained sequences are 0,0,0. The selected times omit both nonzero samples from each signal.

Repair a missing building block

Review Pictures Become Numbers (ai-1/5). Goal: Encode and reconstruct a small image using an explicit grid convention. Short practice: Find a position where the original badges differ, then read the changed A back as a picture.. Return to this week and explain what you changed.

Review Sensors Are Messengers (ai-1/3). Goal: Describe a measurement using its quantity, value, unit and conditions. Short practice: Why can we calculate the earlier gap but not predict an exact new B reading?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
